

Indoor 3D Gaussian Splatting with Structure Awareness for Panorama-Style Mobile Captures

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Abstract

Recent developments in 3D Gaussian Splatting have achieved real-time novel view synthesis with remarkable visual quality. However, producing high-fidelity renderings still demands carefully captured images that comprehensively cover the scene, which restricts usability for non-expert users. While capturing a scene with a single mobile device is convenient, it has been limited in its ability to reliably reconstruct indoor environments due to the fact that its rotation-dominant motion and narrow baseline make accurate camera pose and 3D point estimation challenging, especially in textureless indoor scenes. To address these challenges, we propose a novel and efficient initialization method inspired by man-made structures, which leverages planar structural cues in indoor environments. We also jointly update color inconsistencies and inaccurate camera poses caused by auto-exposure and motion drift on mobile devices. Experimental results show photorealistic rendering, outperforming state-of-the-art methods on the collected real-world scenes.

1. Introduction

Photorealistic novel view synthesis (NVS) is essential for bringing real-world experiences into virtual environments, enabling immersive interaction, especially in indoor scenes, where we spend most of our daily lives. 3D Gaussian Splatting (3DGS) [4] has recently emerged as a powerful framework for real-time NVS, delivering high-fidelity rendering with impressive efficiency. However, achieving high-quality results with 3DGS still requires capturing hundreds of images that densely cover the scene, posing a challenge for general users (i.e., non-experts).

To naturally capture their surroundings, people tend to use handheld devices, often rotating the camera. This casual

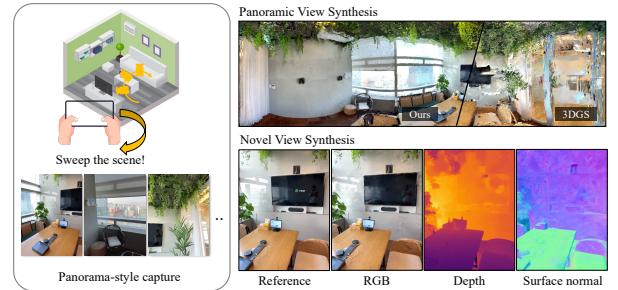


Figure 1. **Overview.** Given consecutive images from panorama-style motion captures, our framework could reconstruct the indoor scenes under ambiguous motion.

motion feels intuitive and requires minimal effort, making it a user-friendly way to scan indoor spaces. Furthermore, this approach supports immersive scene exploration, which is especially useful for AR/VR applications.

In this work, we investigate a method for generating photorealistic NVS from casual captures of indoor scenes, with a focus on improving accessibility for non-expert users. Unlike previous works [5, 8, 10], which rely on a multi-camera rig system composed of high-end cameras to capture 360° scenes, we use only a single mobile device on hand. However, video taken with a casual motion presents additional technical challenges for NVS. Specifically, consecutive images captured with a narrow baseline lead to inaccurate camera pose estimation and unreliable 3D points. These issues are exacerbated in non-textured regions of indoor scenes, where the scarcity of distinct features and limited motion parallax make Structure-from-Motion [6] prone to failure.

To address these issues, we leverage rough geometric priors that are either readily available or estimable. Camera poses can be easily extracted from mobile devices, and monocular depth information can be estimated with reason-

able accuracy using pre-trained networks [12]. It is important to note that these rough geometric priors may be imprecise. Mobile devices relying on IMU sensors for pose estimation are prone to motion drift, and monocular depth estimation methods suffer from scale ambiguity. Nevertheless, these geometric priors can be used with the planar characteristics of the indoor scenes to derive better estimations.

Given rough geometric priors, we first present a new initialization scheme, planar scaffold assembly, that exploits the planar structure of indoor scenes to adjust misaligned depths to enforce global and local consistency to enhance alignments and maintain local details. This approach provides reliable initial values, allowing 3D Gaussians to be aligned with scene geometry. Furthermore, we newly introduce a stable pruning scheme that keeps confident Gaussians that have high opacity values in non-textured regions, enhancing geometric quality and optimization stability. In addition, we corrects initial poses and color inconsistencies using photometric error. By effectively overcoming challenges from casual captures in indoor scenes, our framework achieves photorealistic rendering quality.

1.1. 3D Gaussian Splatting

3D Gaussian Splatting (3DGS) [4] explicitly represents the scene as a set of 3D Gaussian primitives, enabling real-time rendering via differentiable rasterization. Each Gaussian is parameterized as a covariance matrix $\Sigma \in \mathbb{R}^{3 \times 3}$ and a mean $\mu \in \mathbb{R}^3$ in the world space:

$$G(x) = e^{-\frac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu)}, \quad (1)$$

where $\Sigma = RSS^\top R^\top$ is decomposed into a rotation matrix $R \in SO(3)$ and scale matrix $S \in \mathbb{R}^{3 \times 3}$, which is independently stored as a quaternion q and scale factor s , respectively. To project 3D Gaussians into the camera space, the covariance matrix is transformed into the camera space through local affine approximation [14].

Each Gaussian has learnable parameters $(\mu_i, q_i, s_i, \alpha_i, SH_i)$, where α and SH indicate the opacity and spherical harmonics (SH) coefficients, respectively. During α -blending stage, learnable opacity value is multiplied with its projected Gaussian, and a rendered color C is computed as:

$$C = \sum_i^N c_i \alpha_i T_i, \quad (2)$$

where $T_i = \prod_{j=1}^{i-1} (1 - \alpha_j)$ is the transmittance, c_i is attained from the SH basis functions, and N is the number of ordered Gaussians in the overlapped regions with the pixels.

For training 3DGS, a sparse point cloud estimated by COLMAP [6] is used to initialize the locations of a set of 3D Gaussians in the world space. Then, the Gaussian parameters are optimized to minimize rendering loss computed via

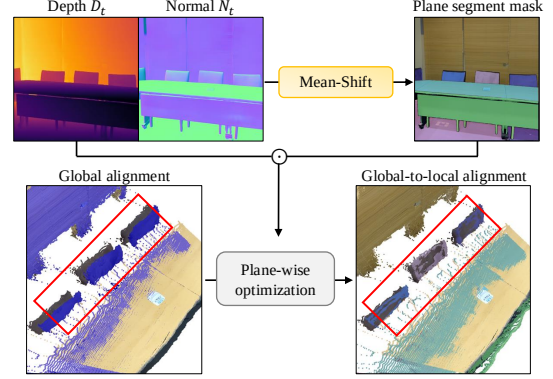


Figure 2. **Effect of plane-wise local alignment.** The red box in the global alignment result shows the local inconsistencies of globally aligned points (blue points). We fine-tune the learned global scale and shift values with the plane-wise local alignment.

differentiable rasterization. Furthermore, 3DGS employs an adaptive density control that clones and splits Gaussians to better represent the fine details of the scene.

2. Indoor 3DGS with Structure Awareness

2.1. Plane Scaffold

Panorama-style motion [7, 9] usually fails to perform COLMAP [6], which estimates accurate camera poses and a sparse point cloud. To resolve this issue, we propose a new alignment scheme, *plane scaffold assembly*, that constructs globally aligned 3D points $\mathcal{X}_T$ and their surface normals $\mathcal{N}_T$, leveraging planar structures from the rough geometric priors (*i.e.*, monocular depth estimation and ARKit poses).

Image-wise Global Alignment. Given previous global points $\mathcal{X}_{t-1}$ and a new depth image D_t , image-wise global alignment aims to predict an affine transformation that adjusts a given depth image to fit global points. Initial global points $\mathcal{X}_0$ are generated by back-projecting the estimated depth from the first frame D_0 .

We represent an adjusted depth map by affine transformation as $\bar{D}_t = \alpha_t D_t + \beta_t$ with the learnable per-view scale and shift parameters $(\alpha_t, \beta_t) \in \mathbb{R}$. These parameters are optimized toward minimizing error between the current depth and projected global points in a gradient descent manner:

$$\underset{\alpha_t, \beta_t}{\operatorname{argmin}} |M \odot (\bar{D}_t - \pi(\mathcal{X}_{t-1}; \Pi_t))|_1, \quad (3)$$

where M is a mask for the overlapped regions between the monocular depth $\bar{D}_t$ and the projected depth, $\odot$ indicates element-wise multiplication to adopt observed pixels during optimization, and $\pi(\cdot; \cdot)$ denotes a projection function by camera pose Π_t . While this alignment scheme yields overall improvements, relying on a single scale and shift parameter to adjust 3D space for various objects causes local inconsistency (see Fig. 2).

Plane-wise Local Alignment. We optimize per-plane scale and shift values to enforce local consistency on globally aligned 3D points. Plane representation gives an important cue for compactly representing the indoor scene, including large homogeneous regions like walls and ceilings. Our intuition is to leverage the geometrical property of the plane in 3D space, where depth values with similar normal directions in local regions can be grouped into the same plane. We adopt efficient mean-shift clustering [13] that groups depth points into separate planes using surface normal embeddings [2], even when the exact number of planes is unknown. Each planar region is represented as a form of a binary mask $m_t^p \in \mathbb{R}^{H \times W}$. Then, we define the per-plane scale and shift parameters (γ_t^p, δ_t^p) for each plane p . Initial values are initialized with the global parameters (α_t, β_t) . The scaled depth $\bar{D}_t^p$ for each plane p is computed by multiplying the plane mask as follows:

$$\bar{D}_t^p = m_t^p \odot (\gamma_t^p D_t^p + \delta_t^p). \quad (4)$$

These local plane regions are optimized while minimizing projection loss between each plane segment and the projected depth within the overlapped pixels as follows:

$$\underset{\gamma_t^p, \delta_t^p}{\operatorname{argmin}} |M \odot (\bar{D}_t^p - \pi(\mathcal{X}_{t-1}; \Pi_t))|_1, \quad (5)$$

where M denotes a mask for the overlapped pixels between the scaled plane depth $\bar{D}_t^p$ and the projected depth. Finally, we back-project adjusted plane segments to obtain 3D points in the world space. They are merged to the global points $\mathcal{X}_{t-1}$ except for overlapped pixels as follows:

$$\mathcal{X}_t = \mathcal{X}_{t-1} \cup \pi^{-1}(\bar{D}_t^p; \Pi_t), \quad (6)$$

where $\pi^{-1}(\cdot; \cdot)$ denotes a back-projection function that back-projects scaled planes into the world space using camera pose. Thanks to the local plane-wise alignment, we observe that 3D points are correctly aligned across various objects or planar regions. In addition, each 3D point has a normal direction belonging to its plane, aggregating into a set of normal vectors $\mathcal{N}_T$.

2.2. Plane-aware Stable Optimization

We introduce a plane-guided initialization, pruning, and regularization that allows us to achieve high-fidelity view synthesis while reducing geometric artifacts. Furthermore, color correction and residual pose refinement resolve inherent issues in mobile captures, such as auto-exposure and motion drift.

Initialization. Given a set of globally aligned 3D points and their surface normals (*i.e.*, plane scaffold), we initialize 3D Gaussian primitives to be aligned with the scene geometry. Specifically, a set of Gaussians is initialized with the positions of the global 3D points. Following 3DGS [4], the

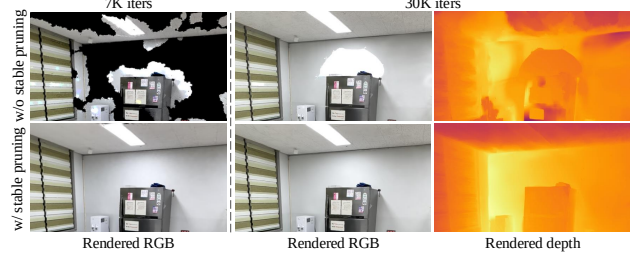


Figure 3. Effect of stable pruning. Unlike previous pruning schemes, stable pruning keeps highly confident Gaussians, preserving precise scene geometry in textureless areas.

initial scale of each Gaussian is determined by the average distance to its nearest neighbors, resulting in an isotropic shape. Thus, we compress the initialized 3D Gaussians by minimizing the scale of the axis closest to the surface normal. By assigning the minimal value to the scale, the 3D Gaussians are initialized with thin structures flattened along the corresponding surface plane.

Stable pruning. The existing pruning scheme removes oversized and transparent Gaussians, frequently observed in non-textured regions. However, this approach degrades optimization stability since the empty holes are filled inappropriately, causing geometric artifacts after optimization. To solve this problem, we present a stable pruning that preserves reliable 3D Gaussians from being pruned in non-textured regions. Intuitively, 3D Gaussians aligned with the surface possess high opacity values, significantly affecting rendering quality. Motivated by this fact, we assess the reliability of 3D Gaussians based on opacity value, as they are initialized from the surface-aligned plane scaffold assembly. Specifically, we first collect pruning candidates for oversized Gaussians within the overlapping regions in the image domain. We then retain high-confidence Gaussians among the candidates whose opacity value exceeds a threshold of 0.5. This simple yet effective strategy improves optimization stability by retaining large Gaussians, thereby preventing large holes in non-textured regions (see Fig. 3).

Regularization. While the 3D Gaussians are initialized with the proposed plane scaffold, relying solely on the rendering loss does not guarantee their geometric accuracy during optimization. Thus, we apply additional loss functions to align 3D Gaussians with scene geometry by computing the loss between estimated rough geometric priors and their corresponding rendered depth and surface normal maps.

Color Correction. Captured images from mobile devices are typically subject to auto-exposure and white balance, leading to color inconsistencies across different view-points. Varying colors across different views are corrected via a simple color transformation strategy parameterized by learnable white balancing $w \in \mathbb{R}^3$ and brightness $b \in \mathbb{R}^3$ as

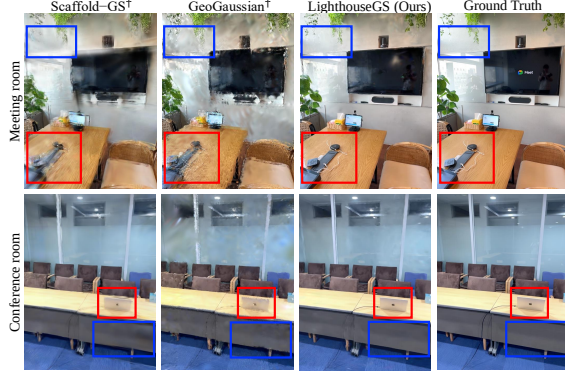


Figure 4. **Qualitative comparisons on the real-world scenes.** † indicates methods that initialize with plane scaffold assembly.

channel-wise parameters. A rendered image $\hat{I}_t$ can be transformed into the color corrected image $\tilde{I}_t$ before calculating rendering loss:

$$\tilde{I}_t = w_t \hat{I}_t + b_t. \quad (7)$$

The learnable coefficients are updated via differentiable rasterization, backpropagating gradients into these parameters.

Residual Pose Refinement. We present a residual pose refinement strategy to correct inaccurate input camera poses caused by motion drift. Inspired by PoRF [3], we optimize residual pose instead of directly refining camera pose parameters. We initialize residual pose parameters for each frame as an identity rotation matrix (quaternion form) and zero translation vector. During optimizing 3DGS, a quaternion and translation vector are transformed into the residual pose $\Delta\Pi \in SE(3)$. To obtain an adjusted pose, the residual pose is multiplied with the initial pose while avoiding direct optimization of the rough initial pose Π as follows:

$$\tilde{\Pi}_t = \Delta\Pi_t \cdot \Pi_t, \quad (8)$$

where $\tilde{\Pi}_t$ indicates the adjusted pose with the learnable residual parameters, but the camera intrinsic is fixed. Then, we compare the rendered image utilizing the adjusted pose with the ground truth image.

3. Dataset and Experiments

Additional results are attached in the supplementary video. **Datasets.** Due to the absence of indoor scene datasets captured by panorama-style motion, we construct a new dataset in real-world scenes to validate the efficiency of our practical scenarios. We use Nerfcapture [1], which saves a set of images with the camera parameters from iPhone’s ARKit. We rotate the device with outstretched half-arms while capturing the region of interest. A set of images covers the almost entire space from the motion center.

Experiments. Figure 4 shows novel view synthesis results on the real-world scenes. We conduct the module ablation study to demonstrate its effectiveness compared to the

Table 1. **Module ablation.** We ablate our three key components on the *meeting room*. The best scores are highlighted in **bold**.

Methods	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	# of Gaussians
3DGS	18.52	0.576	0.483	3.31M
w/o residual pose	20.62	0.604	0.445	2.95M
w/o color correction	20.69	0.675	0.433	2.87M
w/o stable optimization	23.27	0.709	0.353	2.42M
Ours (full model)	23.31	0.719	0.367	2.3M

Table 2. **Initialization ablation.** We compare different initialization algorithms with 3DGS [4] in the *conference room*.

Methods	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
COLMAP	N/A	N/A	N/A
Spherical SfM [9]	17.41	0.660	0.440
ARKit + COLMAP [6]	17.72	0.654	0.443
ARKit + Plane Scaffold (Ours)	23.52	0.789	0.300

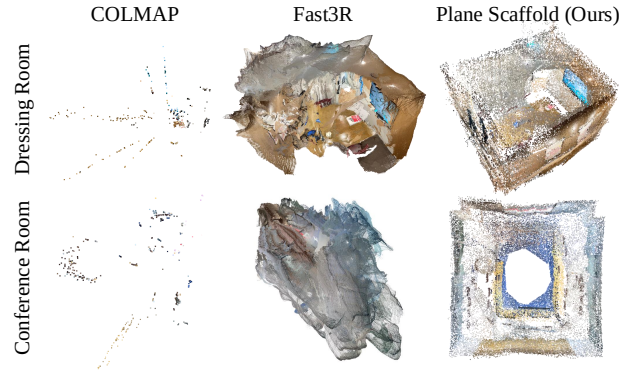


Figure 5. Qualitative comparisons of initial point clouds.

baseline (see Table 1). For the baseline, we initialize 3D Gaussians from sparse point clouds reconstructed by running COLMAP on ARKit poses. Each module contributes to performance gains across all metrics. In Table 2, we compare plane scaffold initialization to conventional SfM algorithms, achieving superior performance across all metrics. Also, Fig. 5 shows initial point clouds generated by various methods, including the geometric foundation model Fast3R [11].

4. Conclusion

We propose a practical 3DGS-based novel view synthesis framework for indoor scenes from panorama-style motion capture with a single mobile device. To address the inherent challenges of motion and indoor scenes, we incorporate planar structures into the initialization, optimization, and regularization. We further correct inaccurate camera poses and auto-exposure images to enhance visual quality. As a result, our method outperforms a photorealistic novel view synthesis with real-time speed while retaining precise scene geometry.

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